

Jaeseung Heo

Graduate School of Artificial Intelligence
POSTECH (Pohang University of Science and Technology)

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EDUCATION

POSTECH

Integrated M.S./Ph.D. Program, Graduate School of Artificial Intelligence
Advisor: Prof. Dongwoo Kim

Pohang, South Korea

Mar, 2021 - Present

Expected Graduation: Feb, 2027

Sungkyunkwan University

B.S. in Computer Science and Engineering
B.S. in Systems Management Engineering (Double Major)

Suwon, South Korea

Mar, 2016 - Aug, 2020

RESEARCH INTERESTS

My research aims to identify which training data causes specific model behaviors, with a longer-term goal of extending this to problematic behaviors relevant to safety and alignment. I use training data attribution (TDA), particularly influence functions, to characterize how training examples drive model behavior, and translate these signals into data-centric interventions such as augmentation, label smoothing, and selection. Methodologically, I develop influence functions that capture dependence between training examples, both explicit (as in graph neural networks) and implicit (arising from joint loss minimization). Going forward, I aim to connect TDA with mechanistic interpretability, and ultimately to trace behaviors such as subliminal learning back to their origins in training data.

PUBLICATIONS

Conference and Workshop Publications

- **Interaction-Aware Influence Functions for Group Attribution** *ICML '26*
Jaeseung Heo, Kyeongheung Yun, Youngbin Choi, Sehyun Hwang, Jungseul Ok, Dongwoo Kim *MI Workshop*
- **Posterior Label Smoothing for Node Classification** *AAAI '26*
Jaeseung Heo, MoonJeong Park, Dongwoo Kim *Oral Pres.*
- **Transductive Generalization and Its Application to Graph Node Classification** *NeurIPS '26*
MoonJeong Park*, Seungbeom Lee*, Kyungmin Kim, Jaeseung Heo, Seunghyuk Cho, Shouheng Li, Sangdon Park, Dongwoo Kim
- **Long Live the Librarian! A Persistent Search Sub-Agent for Energy-Efficient Multi-Agent Software Engineering Systems** *EMNLP '26*
Seunghyuk Cho, Sunghyun Choi, Jaeseung Heo, Youngbin Choi, Saemi Moon, MoonJeong Park, Dongwoo Kim *Oral Pres.*
- **Influence Functions for Edge Edits in Non-Convex Graph Neural Networks** *NeurIPS '25*
Jaeseung Heo, Kyeongheung Yun, Seokwon Yoon, MoonJeong Park, Jungseul Ok, Dongwoo Kim
- **EPIC: Graph Augmentation with Edit Path Interpolation via Learnable Cost** *IJCAI '24*
Jaeseung Heo*, Seungbeom Lee*, Sungsoo Ahn, Dongwoo Kim
- **Mitigating Oversmoothing through Reverse Process of GNNs for Heterophilic Graphs** *ICML '24*
MoonJeong Park, Jaeseung Heo, Dongwoo Kim

Preprints

- **The Oversmoothing Fallacy: A Misguided Narrative in GNN Research** *arXiv '25*
MoonJeong Park, Sunghyun Choi, Jaeseung Heo, Eunhyeok Park, Dongwoo Kim

*Equal contribution.

HONORS AND AWARDS

- **BK21 Outstanding Research Paper Award**, POSTECH *2025, 2026*
- **Dean's List**, Sungkyunkwan University *2018, 2019, 2020*

SERVICE AND TEACHING

- **Reviewer**, ICML, ICML MI Workshop, NeurIPS
- **Teaching Assistant**, CSED700H: Convex Optimization, POSTECH *Fall 2022*
- **Lecturer**, Youth AI & Big Data Academy, AI Expert Training Course, POSCO *2021, 2022*

REFERENCES

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